

(1001CJA102120052)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

FULL SYLLABUS

18-02-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)**ANSWER KEY****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	B	B	A	A	D	D	A	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	B	B	A	D	A	C	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.25	0.40	1.98	1.50	0.32 to 0.35	0.84 to 0.89	40.00	2.50	4.00	2.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	D	A	D	D	A	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	C	D	D	A	C	A	A	B	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	0.00	18.16	3.00	4.00	2.25	4.00	2.00	4.00	20.00	4.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	B	B	C	C	B	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	D	B	B	D	B	C	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	28.00	0.85 or 0.86	77.00	0.82 or 0.83	0.87 or 0.88	4.00	1.00	1.00	2.00	8.00

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-I)
PART-1 : PHYSICS
SOLUTION
SECTION-I

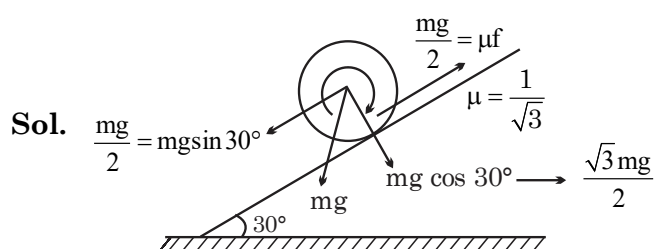
 1. **Ans. (A)**
Sol. $r_n = 1.00 \times 10^{-5} \text{ m},$

$$r_n = 0.01 \text{ mm}$$

$$r_n = (0.525 \times 10^{-10}) \frac{n^2}{z}$$

$$\Rightarrow \sqrt{\frac{0.01 \times 10^{-3}}{0.5 \times 10^{-10}}} = n \Rightarrow \sqrt{\frac{10 \times 10^{-6}}{5.25 \times 10^{-11}}} = n$$

$$n = 435$$

 2. **Ans. (A)**


$$a_{\text{com}} = 0$$

$$\tau_{\text{com}} = \frac{mg}{2} R$$

$$I\alpha = \frac{mg}{2} R$$

$$\Rightarrow \frac{mR^2}{2} \alpha = \frac{mgR}{2} \Rightarrow \alpha = \frac{g}{R}$$

α and ω opposite so rotation stop after sometime and then go downward.

 3. **Ans. (B)**
Sol. Given: $\lambda_1 = \frac{90}{175} \text{ m}$ and $\lambda_2 = \frac{90}{173} \text{ m}$

If f_1 and f_2 are the corresponding frequencies and v is the velocity of sound in air, we have

$$v = f_1 \lambda_1 \text{ and } v = f_2 \lambda_2$$

$$f_1 = \frac{v}{\lambda_1} \text{ and } f_2 = \frac{v}{\lambda_2}$$

Since, $\lambda_1 < \lambda_2$, we must have $f_1 > f_2$

If f is the frequency of the third note, then

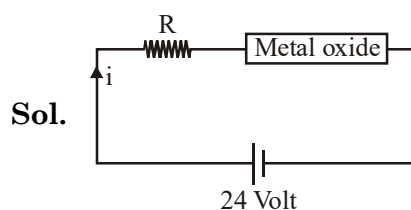
$$f_1 - f = 4 \text{ and } f - f_2 = 4$$

$$\Rightarrow f_1 - f_2 = 8$$

$$\frac{v}{\lambda_1} - \frac{v}{\lambda_2} = 8$$

$$v \left[\frac{175}{90} - \frac{173}{90} \right] = 8$$

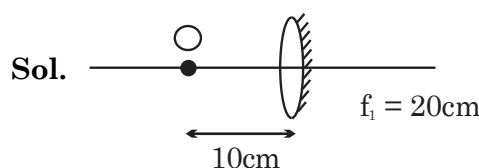
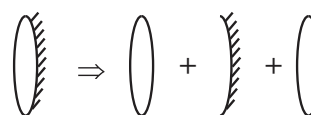
$$v = 360 \text{ m/s.}$$

 4. **Ans. (B)**

Sol.

Applying Kirchoff's Law

$$24 - iR - Ki^2 = 0$$

$$\text{and } P_{\text{rod}} = 2P_R$$

 5. **Ans. (A)**

Sol.


It behave like concave mirror

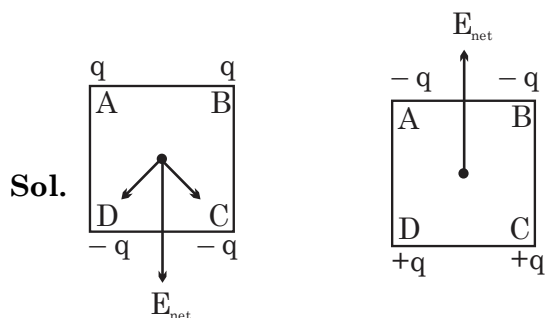
$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_m} + \frac{1}{f_1} = \frac{2}{f_1} = \frac{1}{f_m}$$

$$= \frac{2}{20} + \frac{1}{22/2} = \frac{21}{110} \text{ cm}$$

$$\frac{1}{v} + \frac{1}{-10} = -\frac{1}{110/21}$$

$$\Rightarrow v = -11 \text{ cm real image.}$$

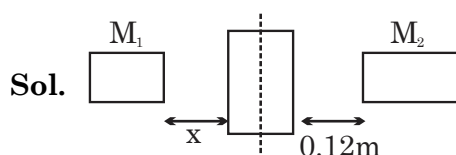
6. Ans. (A)



direction of field changes.

but potential is scalar addition due to all charge it remain same.

7. Ans. (D)



$$\frac{M_1}{M_2} = \frac{27}{8}$$

$$B_1 = \frac{\mu_0 (2M_1)}{4\pi x^3}; \quad B_2 = \frac{\mu_0 (2M_2)}{4\pi (0.12)^3}$$

$$\frac{27}{8} = \frac{x^3}{(0.12)^3} \Rightarrow x = 0.18\text{m}$$

8. Ans. (D)

Sol. $t = 0$

capacitor behaves like plane wire

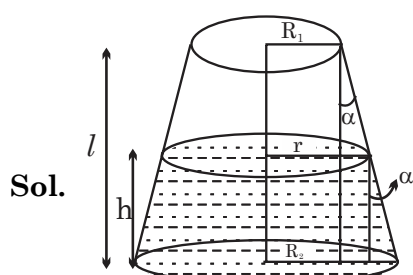
$$\therefore i = \frac{2}{1000} = 2\text{mA}$$

$t = \infty$

capacitor behave like open circuit

$$i = \frac{2}{2000} \text{A} = 1\text{mA}$$

9. Ans. (A)



$$\frac{R_2 - r}{h} = \frac{R_2 - R_1}{\ell} = \tan \alpha$$

$$\Rightarrow r = R_1 - (R_1 - R_2) \frac{h}{\ell}$$

$$\Rightarrow r = 0.5 \times 10^{-3} - (0.5 \times 10^{-3} - 2.5 \times 10^{-4}) \frac{8 \times 10^{-2}}{10^{-1}}$$

$$= 0.5 \times 10^{-3} - (0.25 \times 10^{-3}) \times 8 \times 10^{-1}$$

$$= 0.5 \times 10^{-3} - 2 \times 10^{-4}$$

$$= 0.3 \times 10^{-3} \text{ m}$$

Surface tension of liquid at 0°C

$$S_0 = \frac{r h \rho g}{2 \cos \theta} \quad (\theta = 0)$$

$$= \frac{0.3 \times 10^{-3} \times 8 \times 10^{-2} \times \frac{10^4}{14} \times 9.8}{2}$$

$$= \frac{0.3 \times 8 \times 9.8 \times 10^{-1}}{2 \times 14} = \frac{0.6 \times 9.8}{7} \times 10^{-1}$$

$$= 0.084 \text{ N/m}$$

If $r_0 \rightarrow$ radius of cylindrical tube

$$S_0 = \frac{r_0 h \rho g}{2} \quad (\text{at } 0^\circ\text{C})$$

$$S_{50} = \frac{r_0 h_{50} \rho g}{2} \quad (\text{at } 50^\circ\text{C})$$

$$\frac{S_{50}}{S_0} = \frac{h_{50}}{h_0} = \frac{5.5 \times 10^{-2}}{6 \times 10^{-2}} = \frac{11}{12}$$

$$S_{50} = \frac{11}{12} \times S_0 = \frac{11}{12} \times 0.084 = 0.077 \text{ N/m}$$

Rate of change of surface tension with temperature

$$\frac{S_{50} - S_0}{50} = \frac{0.077 - 0.084}{50} = -1.4 \times 10^{-4} \frac{\text{N}}{\text{m}^\circ\text{C}}$$

10. Ans. (D)

Sol. $I = \frac{I_0}{2} \cos^2 \theta$

after first polaroid

$$\frac{I}{I_0} = \frac{1}{2} \cos 2\theta$$

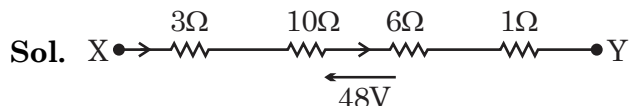
after 6th

$$\frac{I}{I_0} = \frac{1}{2} (\cos 2\theta)^5 = \frac{1}{2} \left(\frac{\sqrt{3}}{2} \right)^{10}$$

$$= \frac{81 \times 3}{2 \times 32 \times 32} = 11.8$$

$$\% \approx 12\%$$

11. Ans. (A)

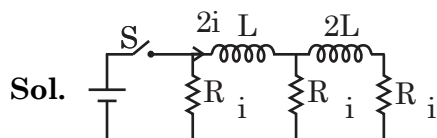


$$i = \frac{48}{6} = 8$$

$$R_{\text{net}} = 20\Omega$$

$$V_{xy} = 160 \text{ volt}$$

12. Ans. (A)

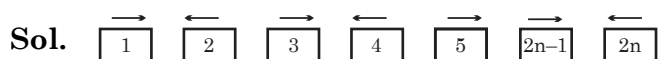


$$\frac{1}{2}L(2i)^2 + \frac{1}{2}2Li^2$$

$$\Rightarrow 2Li^2 + Li^2 = 3Li^2$$

13. Ans. (B)

14. Ans. (B)



1st → n collision

velocity exchange

2nd → n - 1 collision

n - 2

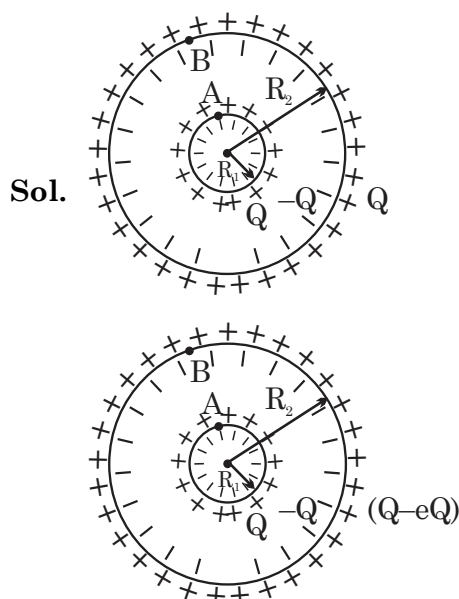
⋮

1

$$n + (n - 1) + (n - 2) + \dots + 1$$

$$= \frac{n(n+1)}{2}$$

15. Ans. (A)



$$V_A = \frac{KQ}{R_1} + \frac{K(-Q)}{R_2} + \frac{K(+Q)}{R_2} = \frac{KQ}{R_1}$$

$$V_B = \frac{KQ}{R_2} + \frac{K(-Q)}{R_2} + \frac{K(+Q)}{R_2} = \frac{KQ}{R_2}$$

$$\Delta V = V = V_B - V_A = KQ \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$V_A = \frac{KQ}{R_1} + \frac{K(-Q)}{R_2} + \frac{K(Q - eQ)}{R_2}$$

$$= \frac{KQ}{R_1} - \frac{KeQ}{R_2}$$

$$V_B = \frac{KQ}{R_2} - \frac{K(-Q)}{R_2} + \frac{K(Q - eQ)}{R_2}$$

$$= \frac{KQ}{R_2} - \frac{KeQ}{R_2}$$

$$\Delta V = V_A - V_B = \frac{KQ}{R_1} - \frac{KQ}{R_2}$$

same potential difference.

16. Ans. (D)

Sol. $\mu_{\text{total}} = N\mu$ (N is no. of iron atoms)

$m = Nm$ (m is mass of iron sphere)

$$M = PV$$

$$\mu = PV$$

$$\Rightarrow Nm = P$$

$$\Rightarrow N = \frac{PV}{m}$$

$$\mu_{\text{total}} = Nm = \frac{PV}{m} \mu$$

$$V_S = \left(\frac{m\mu_{\text{total}}}{P\mu} \right)$$

$$\text{fraction} = \frac{V_S}{V_E} = \left(\frac{m\mu_{\text{total}}}{P\mu \frac{4}{3}\pi R_e^3} \right)$$

$$\frac{V_S}{V_E} = \frac{9.3 \times 10^{-26} \times 8 \times 10^{22} \times 3}{(14 \times 10^3)(2.1 \times 10^{-23})4 \times 3.14 \times (6.4 \times 10^6)^3}$$

$$= \frac{9.3 \times 8 \times 3 \times 10^{-4}}{14 \times 2.1 \times 4 \times 3.14 \times (6.4)^3 \times 10^{-2}}$$

$$= \frac{9.3 \times 8 \times 3 \times 10^{-2}}{14 \times 2.1 \times 4 \times 3.14 \times (6.4)^3}$$

$$= \frac{223.2 \times 10^{-2}}{96800.34} = 2.3 \times 10^{-5}$$

$$\text{mass of iron atom} = 560 \times 1.66 \times 10^{-27} \text{ kg}$$

$$= 9.3 \times 10^{-26} \text{ kg}$$

17. Ans. (A)

Sol. $\frac{V}{\sqrt{R^2 + X_C^2}} = I; \frac{V}{\sqrt{R^2 + (3X_C)^2}} = \frac{I}{2}$

Take ratio

$$(2)^2 = \frac{R^2 + 9X_C^2}{R^2 + X_C^2}$$

$$\Rightarrow 4R^2 + 4X_C^2 = R^2 + 9X_C^2$$

$$\Rightarrow 3R^2 = 5X_C^2$$

18. Ans. (C)

Sol. Pressure of reflection = $\frac{2I}{C} = \frac{2 \text{ Power}}{A C}$

$$= \frac{2 \times 1.5 \times 10^9}{1.3 \times 10^{-6} \times 3 \times 10^8} = \frac{2 \times 1.5}{1.3 \times 3} \times 10^7 = \frac{1}{1.3} \times 10^7$$

$$= 7.7 \times 10^6$$

19. Ans. (A)

Sol. Least count = $\frac{\text{smallest reading on main scale}}{\text{No. of division on vernier scale}}$

$$= \frac{a}{n+1}$$

20. Ans. (B)

Sol. $T_1 = \pi \sqrt{\frac{l}{g}}; T_2 = \pi \sqrt{\frac{l'}{g}} = \pi \sqrt{\frac{l}{4g}}$

$$mgl(1 - \cos \theta) = mgl'(1 - \cos 2\theta)$$

$$l(1 - \cos \theta) = l'(1 - \cos 2\theta)$$

$$l(2 \sin^2(\theta/2)) = l'(2 \sin^2 \theta)$$

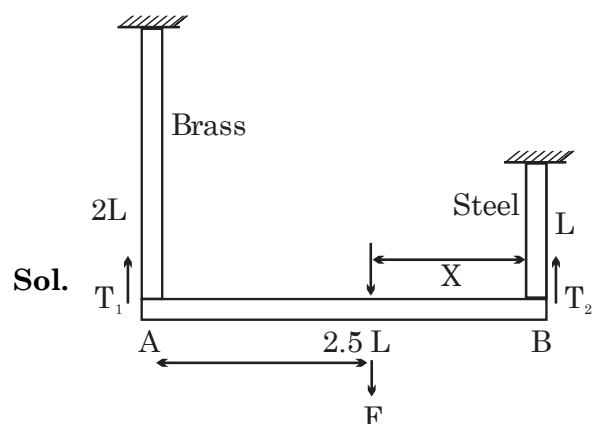
$$\Rightarrow l \sin^2(\theta/2) = l' \sin^2 \theta$$

$$\Rightarrow \sqrt{\frac{l}{l'}} = 2 \Rightarrow l = 4l'$$

$$T_1 + T_2 = \frac{3\pi}{2} \sqrt{\frac{l}{g}}$$

SECTION-II

1. Ans. 1.25



$$\tau_F = 0$$

$$T_2 X = T_1 (2.5L - X)$$

$$\Rightarrow \frac{T_2}{T_1} = \left(2.5 \frac{L}{X} - 1 \right)$$

$$1 = \left(2.5 \frac{L}{X} - 1 \right)$$

$$\eta = 1.25L$$

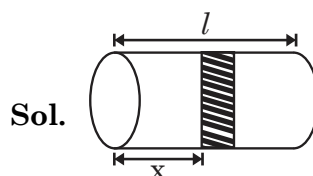
$$Y_E = \sigma$$

$$\Rightarrow T_1 = Y_B \pi 4r^2 \frac{\Delta L}{2L}$$

$$\Rightarrow T_2 = Y_S \pi r^2 \frac{\Delta L}{L}$$

$$\Rightarrow \frac{T_1}{T_2} = \frac{2Y_B}{Y_S} = 1$$

2. Ans. 0.40



Sol.

$$\int dR = \int \frac{dx}{\sigma A} \Rightarrow R = \int_0^l \frac{x dx}{\sigma_0 l A}$$

$$R = \frac{1}{2\sigma_0 A}$$

$$j = \sigma E \Rightarrow E = \frac{j}{\sigma}$$

$$\frac{i}{A\sigma} = \frac{V}{Ra\sigma} = \frac{Vx}{RA\sigma_0 l} = \frac{Vx}{(l/2)l} = \frac{10 \times 2}{(100/2)}$$

3. Ans. 1.98

Sol. $\gamma \rightarrow {}_{-1}e^0 + {}_{+1}e^0$

from energy mass equivalence

$$E = m_e C^2 + M_e C^2$$

$$= 2m_e C^2 = \frac{2 \times 9.1 \times 10^{-31} \times (3 \times 10^8)^2}{1.6 \times 10^{-19} \times 10^6}$$

$$= 1.02 \text{ MeV}$$

$$\therefore \text{K.E.} = 3 - 1.02 = 1.98 \text{ MeV}$$

4. Ans. 1.50

Sol. $\left(\frac{I_p}{I_{\max}} \right)_{\lambda_1} = \cos^2 \frac{\phi}{2} = \frac{75}{100}$

$$\cos \frac{\phi}{2} = \frac{\sqrt{3}}{2}$$

$$\phi = \frac{\pi}{3}$$

$$\Delta x = \frac{\lambda_1}{2\pi} \phi = \frac{\lambda_1}{6}$$

$$\lambda_1 = 6\Delta x$$

$$\left(\frac{I_p}{I_{\max}} \right)_{\lambda_2} = \cos^2 \frac{\phi'}{2} = \frac{50}{100}$$

$$\cos \frac{\phi'}{2} = \frac{1}{\sqrt{2}}$$

$$\phi' = \frac{\pi}{2}$$

$$\Delta x = \frac{\lambda_2}{2\pi} \phi = \frac{\lambda_2}{4}$$

$$\lambda_2 = 4\Delta x$$

$$\lambda_1 : \lambda_2 = 3 : 2 = 1.5$$

5. Ans. 0.32 to 0.35

Sol. $P_1 = \frac{(100)^2}{R}$; $P_2 = \frac{(141)^2}{R}$

$$\frac{P_1}{P_2} = \left(\frac{100}{141} \right)^2 = \left(\frac{1}{1.41} \right)^2 = \frac{1}{2}$$

$$P_1 = \sigma e A (T^4 - T_0^4)$$

$$P_2 = \sigma e \frac{A}{2} (T^4 - T_0^4) + \sigma \frac{A}{2} (T^4 - T_0^4)$$

$$\frac{P_1}{P_2} = \frac{\sigma e A}{\frac{\sigma A}{2}(e+1)} = \frac{2e}{e+1}$$

$$\Rightarrow \frac{1}{2} = \frac{2e}{e+1}$$

$$\Rightarrow eH = 4e$$

$$\Rightarrow e = \frac{1}{3} = 0.33$$

6. Ans. 0.84 to 0.89

Sol. $\frac{F_t}{A} = \eta \phi$

$$\phi = \frac{F_t}{\eta A} = \frac{25}{1.9 \times 10^4 \times 15 \times 10^{-4}} = 0.87$$

7. Ans. 40.00

8. Ans. 2.50

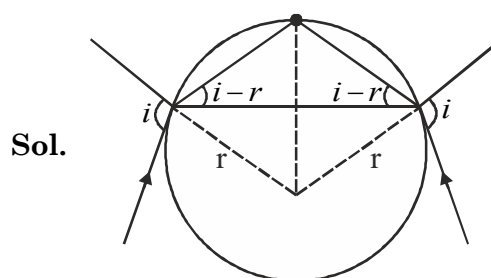
Sol. $P = F_{\text{Thrust}} v$

$$P = (\lambda v)v = \lambda v^2$$

9. Ans. 4.00

Sol. $F = p \frac{dE}{dr} \Rightarrow F = p \frac{\lambda}{2\pi\epsilon_0} \left(-\frac{1}{r^2} \right)$

10. Ans. 2.00



By Geometry, $2R \sin \left(\frac{i}{2} \right) \cos(i-2) = R \cos r$
and $\sin i = \sqrt{3} \sin r$

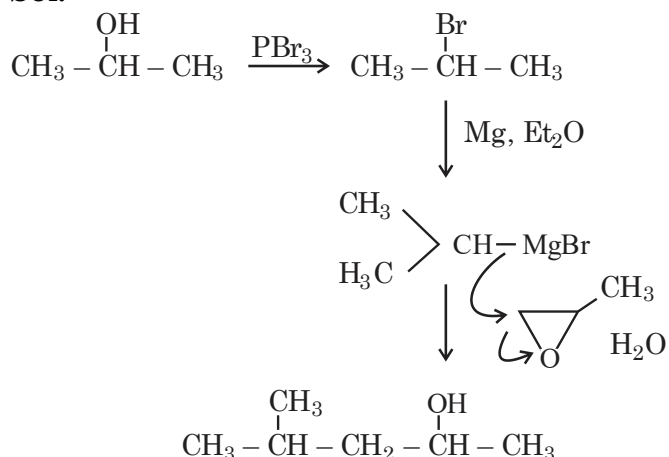
PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A)

Sol.



2. Ans. (B)

Sol. Strength of O-D-bond > O-H bond

3. Ans. (D)

Sol. For ideal gas $\Delta S = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$

$$\text{Given, } n = 1, v_1 = V, V_2 = \frac{V}{2}$$

$$T_1 = T, T_2 = 2T$$

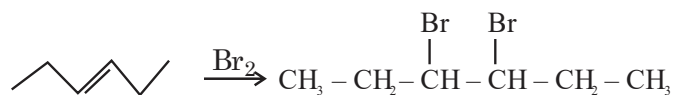
$$\text{So, } \Delta S = 1C_v \ln \frac{2T}{T} + 1R \ln \frac{V/2}{V}$$

$$\Delta S = C_v \ln 2 - R \ln 2$$

$$\Delta S = (C_v - R) \ln 2$$

4. **Ans.(A)**

Sol.



(X)

$$X = 1$$

(Optically inactive)

(due to internal compensation)

$$\text{No. of stereoisomers} = \underset{\substack{\downarrow \\ \text{O.A.}}}{2^1} + \underset{\substack{\downarrow \\ \text{Meso} \\ \text{Comp.} \\ \text{(O.I.A)}}}{2^0}$$

5. Ans. (D)

Sol. NCERT

6. Ans.(D)

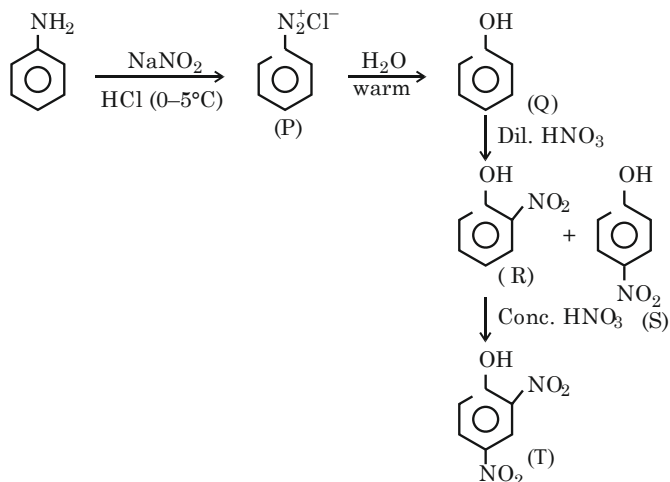
Sol. For rock salt structure,

$$a = 2(r^+ + r^-)$$

$$a = 2(181 + 101) \text{ pm} = 564 \text{ pm}$$

7. Ans. (A)

Sol.



8. **Ans. (A)**

Sol. NCERT

9. Ans. (A)

Sol. At cathode: $Cu_{(aq)}^{+2} + 2e^{-} \longrightarrow Cu_{(s)}$

$$\text{at anode : } 2H_2O_{(l)} \longrightarrow O_{2(g)} + 4H^+_{(aq)} + 4e^-$$

Number of equivalent of Cu = number of
equivalent of O,

$$n \times n_f)_{Cu} = n \times n_f)_{O_s}$$

$$n \times 2 = \frac{5.6}{22.4} \times 4$$

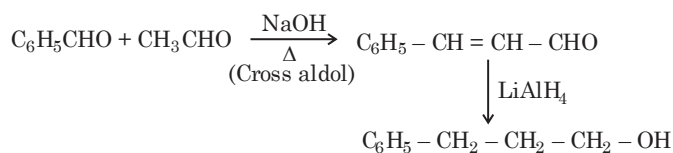
$$n = \frac{1}{2}$$

$$\therefore \text{mass of Cu} = \frac{1}{2} \times 63.5 \text{ gm}$$

$$= 31.75 \text{ gm}$$

10. Ans. (B)

Sol.



11. Ans. (C)

Sol. NCERT

12. Ans. (C)

Sol. Molar heat of vapourisation

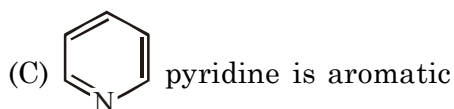
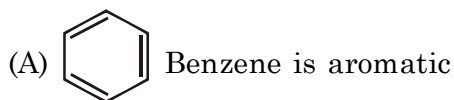
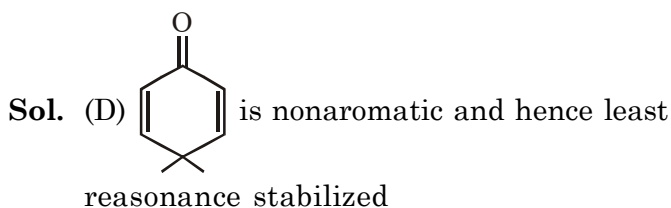
$$= 40.85 \times 10^3 \text{ J mol}^{-1}$$

Heat required to convert 3.6 gm liquid water

$$\text{to water vapour} = \frac{40850}{18} \times 3.6 = 8170 \text{ J}$$

$$\therefore \Delta S = \frac{8170}{373} = 21.89 \text{ JK}^{-1}$$

13. Ans. (D)



14. Ans. (D)

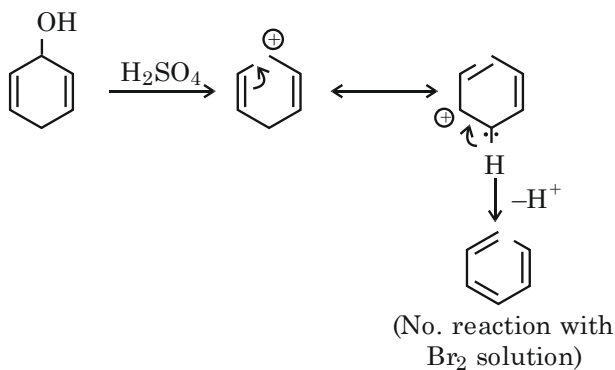
Sol. Wilkinson catalyst – $[\text{RhCl}(\text{PPh}_3)_3] = 16e^-$
 cis-platin – $[\text{PtCl}_2(\text{NH}_3)_2] = 16e^-$
 Brown ring complex – $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4 = 19e^-$
 Prussian blue – $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 = 18e^-$

15. Ans. (A)

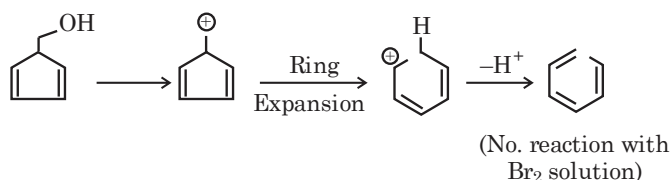
Sol. $\text{HPh} + \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{Ph}^-$
 Colourless Pink
 In alkaline medium, H_3O^+ gets consumed and reaction proceeds in forward direction. As a result pink colour appears as extent of Ph^- increases.

16. Ans. (C)

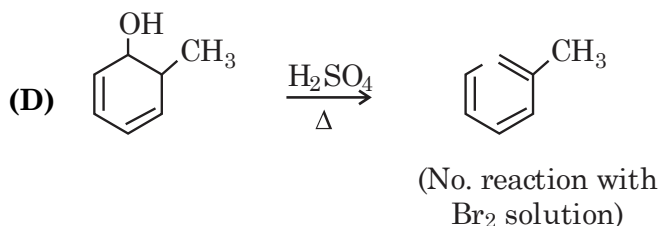
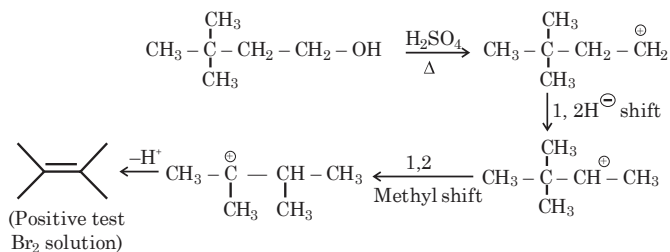
Sol.
 (A)



(B)



(C)

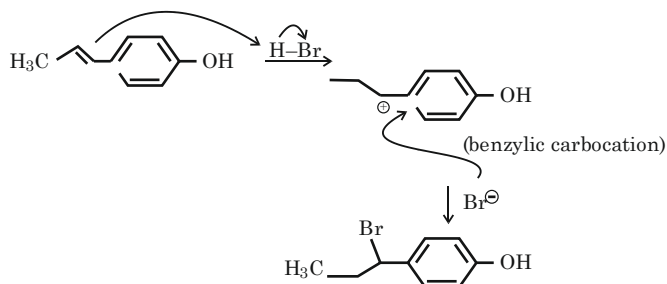


17. Ans. (A)

Sol. (a) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \longrightarrow 6\text{Cu} + \text{SO}_2$
 (b) Al is reduced by electrolytic method
 (c) The green flame in the Bessmer converter is due to the presence of uncombined FeO .

18. Ans. (A)

Sol.



19. Ans. (B)

Sol. $\Delta x \cdot \Delta p = \frac{h}{4\pi}$

$$\lambda = \frac{h}{p}$$

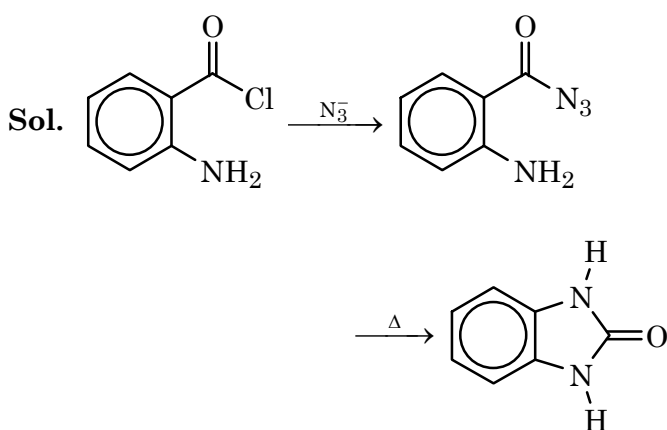
$$\therefore \frac{\Delta \lambda}{\lambda} = \frac{\Delta p}{p}$$

$$\Rightarrow \Delta p = \frac{p \Delta \lambda}{\lambda} = \frac{h \Delta \lambda}{\lambda^2}$$

$$\therefore \Delta x = \frac{h}{4\pi \Delta p} = \frac{h \lambda^2}{4\pi h \Delta \lambda}$$

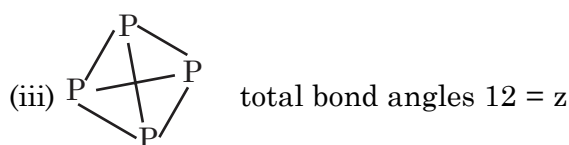
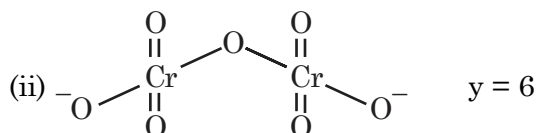
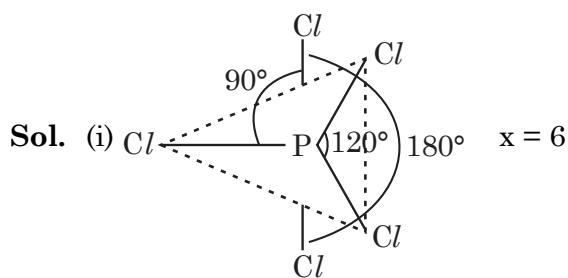
$$= \frac{(5000 \times 10^{-10})^2}{4 \times 3.14 \times 10^{-12}} = 0.02 \text{ m}$$

20. Ans. (C)



SECTION-II

1. Ans. (0.00)



2. Ans. (18.16)

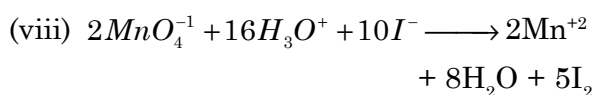
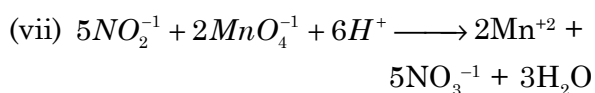
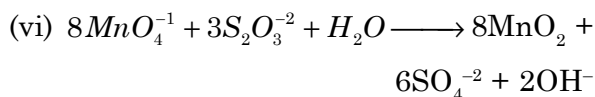
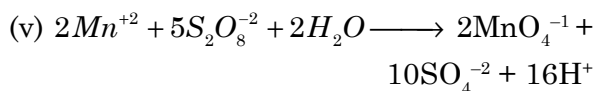
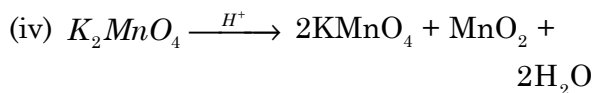
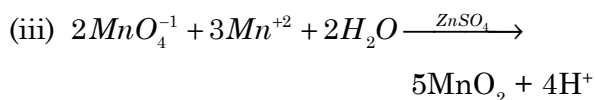
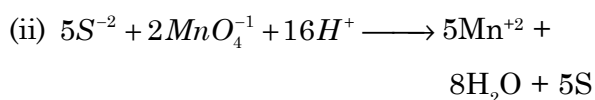
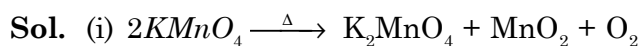
Sol. Molarity = $\frac{\text{Volume strength}}{11.35}$

Volume strength = $11.35 \times 1.6 = 18.16$ V

3. Ans. (3.00)

Sol. 3, 7, 8

4. Ans. (4.00)



5. Ans. (2.25)

Sol. $\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{1}{6} = 1 - \frac{T_2}{T_1} \Rightarrow \frac{T_2}{T_1} = \frac{5}{6}$

$\Rightarrow T_2 = \frac{5}{6}T_1 \dots(1)$

$\eta = 1 - \frac{T_2}{T_1} \Rightarrow \frac{1}{3} = 1 - \frac{T_2 - 65}{T_1} \Rightarrow \frac{T_2 - 65}{T_1} = \frac{2}{3}$

$\Rightarrow T_2 - 65 = \frac{2}{3}T_1 \dots(2)$

by equation (1) & (2)

$\frac{5}{6}T_1 - 65 = \frac{2}{3}T_1$

$T_1 = 390 \text{ K} = 117^\circ\text{C}$

and $T_2 = \frac{5}{6}T_1 = 325 \text{ K} = 52^\circ\text{C}$

$\therefore \frac{T_1}{T_2} = \frac{117}{52} = 2.25$

6. Ans. (4.00)

7. Ans. (2.00)

8. Ans. (4.00)

9. Ans. (20.00)

10. Ans. (4.00)

Sol. 1, 3, 4, 6

SECTION-I

1. Ans. C

Sol. $A = n \left(\frac{a+b}{2} \right) = 18 \left(\frac{9+18}{2} \right) = 3^5$

$$G = (\sqrt{ab})^n = (\sqrt{9 \times 18})^{20} = 2^{10} \cdot 3^{40}$$

Now, $x^2 + 2^6 \cdot 3^{20}x + 2^{10} \cdot 3^{40} = 0$

$$(x + 2^5 \cdot 3^{20})^2 = 0$$

$\Rightarrow$ Roots are equal, real & Negative.

2. Ans. C

Sol. PQ be perpendicular to both lines

$$\Rightarrow \text{for dr's of PQ} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & -1 \\ 8 & 1 & -4 \end{vmatrix}$$

$$= \hat{i}(9) - \hat{j}(0) + \hat{k}(18)$$

$$\Rightarrow \text{dr's of PQ } 9, 0, 18$$

$$\Rightarrow \text{dr's of PQ } 1, 0, 2$$

$$\Rightarrow \text{dr's of PQ } -1, 0, -2$$

3. Ans. (A)

Method 1 :

$$x^2 + 2xy + y^2 - 8x - 8y + 12 = 0$$

$$\Rightarrow (x+y)^2 - 8(x+y) + 12 = 0$$

$$\Rightarrow (x+y)^2 - 6(x+y) - 2(x+y) + 12 = 0$$

$$\Rightarrow (x+y)[x+y-6] - 2[x+y-6] = 0$$

$$\Rightarrow (x+y-2)(x+y-6) = 0$$

$\Rightarrow$ a pair of parallel lines

Method 2 :

here $h^2 = ab$

$$\text{and } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

$\Rightarrow$ a pair of parallel lines.

4. Ans. B

Sol. $r = \sqrt{4+1+20} = 5$

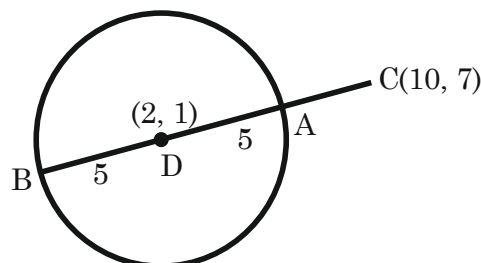
Slope of line

$$DP = \frac{7-1}{10-2} = \frac{3}{4}$$

$$\tan \theta = \frac{3}{4} \Rightarrow \cos \theta = \frac{4}{5} \text{ and } \sin \theta = \frac{3}{5}$$

For point A & B

$$(2 \pm 5 \cos \theta, 1 \pm 5 \sin \theta)$$



$$\left(2 \pm 5 \left(\frac{4}{5} \right), 1 \pm 5 \left(\frac{3}{5} \right) \right)$$

$$(2 \pm 4, 1 \pm 3)$$

$$A(6, 4), B(-2, -2)$$

$$\text{Now } AB = 10$$

$$\Rightarrow PA + PB = 15 > AB$$

$\Rightarrow$ locus of P be an ellipse

$$\Rightarrow 2a = 15, 2ae = 10$$

$$\Rightarrow e = \frac{10}{15} = \frac{2}{3}$$

5. Ans. B

Sol. $A^{-1} = \frac{\text{adj}A}{|A|}$

$$\Rightarrow \left| \text{adj} \left(2 \cdot \frac{\text{adj}A}{|A|} \right) \right| = \left| \text{adj} \left(\frac{2}{|A|} \text{adj}A \right) \right|$$

$$= \left| \frac{4}{|A|^2} \text{adj}(\text{adj}A) \right|$$

$$= \frac{4^3}{|A|^6} \cdot |\text{adj}(\text{adj}A)| = \frac{4^3}{|A|^6} \cdot |A|^4 = \frac{2^6}{|A|^2}$$

$$= \frac{2^6}{|\text{adj}B|^2} = \frac{2^6}{|B|^4} = 4$$

6. Ans. C

Sol. Let $a = b = \sqrt{2} + 1 \Rightarrow a(\sqrt{2} - 1)$

$$\Rightarrow (\sqrt{2} + 1)(\sqrt{2} - 1) = 1 \text{ (Which is not irrational)}$$

$\Rightarrow$ Not reflexive

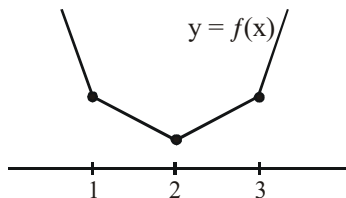
$$(2, \sqrt{2}) \in R_1 \text{ but } (\sqrt{2}, 2) \notin R_1$$

$\Rightarrow$ Not symmetric.

$(2, \sqrt{2}) \in R_1, (\sqrt{2}, 2\sqrt{2}) \in R_1$ but $(2, 2\sqrt{2}) \notin R_1$

$\Rightarrow$ Not transitive

7. **Ans. (C)**



If we draw the graph of $y = f(x)$ then it is clearly seen that sharp corners appear at $x = 1, 2, 3$ so non differentiable at 3 points.

8. **Ans. (B)**

We know that

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2 = {}^{2n}C_n$$

so

$$({}^{12}C_0)^2 + ({}^{12}C_1)^2 + \dots + ({}^{12}C_{12})^2 = {}^{24}C_{12}$$

9. **Ans. (C)**

according to sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

Now if $a = 2$ then $\angle A = \frac{\pi}{6}$

$$\Rightarrow \frac{2}{\sin \frac{\pi}{6}} = 2R \Rightarrow R = 2 \text{ units}$$

10. **Ans. B**

Sol. $\tan 60^\circ = \frac{r}{h}$

$$r = \sqrt{3}h \quad \dots(1)$$

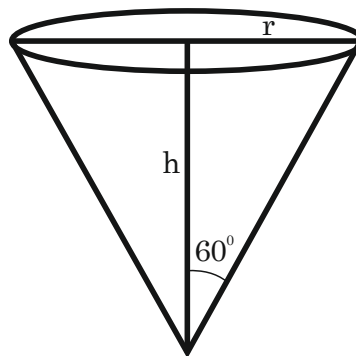
$$\text{Now } \frac{dv}{dt} = 3$$

$$\frac{d}{dt} \left(\frac{1}{3} \pi r^2 h \right) = 3 \quad (\text{put } r = \sqrt{3}h)$$

$$\frac{d}{dt} (\pi h^3) = 3$$

$$3\pi h^2 \frac{dh}{dt} = 3$$

$$\frac{dh}{dt} = \frac{1}{\pi h^2}$$



$$\begin{aligned} \frac{d}{dt} (2\pi r) &= \frac{d}{dt} (2\sqrt{3}\pi h) = 2\sqrt{3}\pi \frac{dh}{dt} \\ &= 2\sqrt{3}\pi \times \frac{1}{\pi h^2} \end{aligned}$$

$$\text{at } h = 1, \frac{d}{dt} (2\pi r) = 2\sqrt{3} \text{ m/s}$$

11. **Ans. C**

Sol. Variance of 201, 203, ..., 299 is same as variance of 2, 4, 6, ..., 100

$$\begin{aligned} \sigma^2 &= \frac{1}{n^2} [n \sum x_i^2 - (\sum x_i)^2] \\ &= \frac{1}{50^2} [50(2^2 + 4^2 + \dots + 100^2) - (2 + 4 + \dots + 100)^2] \\ &= \frac{1}{50^2} [50 \times 4(1^2 + 2^2 + \dots + 50^2) - 4(1 + 2 + \dots + 50)^2] \\ &= \frac{1}{50^2} [2.50^2 \cdot 17.101 - 50^2 \cdot 51^2] \\ &= 17(202 - 153) = 17 \times 49 = 833 \end{aligned}$$

12. **Ans. D**

Sol. $f(x) = 1 + \frac{\int_1^x f(t) dt}{x}$

$$xf(x) = x + \int_1^x f(t) dt$$

$$f(x) + xf'(x) = 1 + f(x)$$

$$f'(x) = \frac{1}{x}$$

$$f(x) = \ln_e |x| + C$$

$$\because f(1) = 1 \Rightarrow f(x) = \ln_e |x| + 1$$

Required area

$$= \int_0^1 e^{y-1} dy = \left(1 - \frac{1}{e} \right)$$

13. Ans. D

Sol. $\begin{vmatrix} \sin 3\theta & \cos 2\theta & 2 \\ 1 & 3 & 7 \\ 1 & -4 & -7 \end{vmatrix} = 0$

$$\sin 3\theta(7) - \cos 2\theta(-14) + 2(-7) = 0$$

$$\sin 3\theta + 2 \cos 2\theta - 2 = 0$$

$$3 \sin \theta - 4 \sin^3 \theta + 2(1 - 2 \sin^2 \theta) - 2 = 0$$

$$3 \sin \theta - 4 \sin^3 \theta - 4 \sin^2 \theta = 0$$

$$\sin \theta(4 \sin^2 \theta + 4 \sin \theta - 3) = 0$$

$$\sin \theta(4 \sin^2 \theta + 6 \sin \theta - 2 \sin \theta - 3) = 0$$

$$\sin \theta(2 \sin \theta(2 \sin \theta + 3) - 1(2 \sin \theta + 3)) = 0$$

$$\sin \theta(2 \sin \theta - 1)(2 \sin \theta + 3) = 0$$

$$\sin \theta = 0, \frac{1}{2}, -\frac{3}{2} \text{ (Reject)}$$

$$\sin \theta = 0, \frac{1}{2}$$

$$\theta = 0, \pi, 2\pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\Rightarrow 5 \text{ solutions}$$

14. Ans. B

Sol. Use $AM \geq GM$, we get $2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}}$
 $\Rightarrow$ solution does exist if $\sin x + \cos x = -\sqrt{2}$

$$\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x = -1$$

$$\cos\left(x - \frac{\pi}{4}\right) = -1$$

$$\cos\left(x - \frac{\pi}{4}\right) = -1 \Rightarrow x - \frac{\pi}{4} = (2n+1)\pi$$

$$\Rightarrow x = (2n+1)\pi + \frac{\pi}{4}$$

$$\Rightarrow \text{Solutions in the interval } [-2\pi, 3\pi] \text{ are}$$

$$\frac{5\pi}{4}, -\frac{3\pi}{4}$$

$$\Rightarrow 2 \text{ solutions.}$$

15. Ans. B

Sol. Let p is $x = 3$

q is $y = 4$

and r is $xy - 9y - 3y + 27 = 0$

contrapositive of $(pvq) \rightarrow r$

is $\neg r \rightarrow \neg(pvq)$

$$\neg r \rightarrow (\neg p \wedge \neg q)$$

if $xy - 9y - 3y + 27 \neq 0$ then $x \neq 3$ and $x \neq 4$

16. Ans. D

Sol. Let $g(x) = \sin^2 x + 4 \operatorname{cosec}^2 x$
 $= \sin^2 x + \cos^2 x + 3 \operatorname{cosec}^2 x$
 $= (\sin x - \operatorname{cosec} x)^2 + 2 + 3 \operatorname{cosec}^2 x$
 $\Rightarrow g(x)$ is minimum when

$$\sin x - \operatorname{cosec} x = 0 \Rightarrow x = \frac{\pi}{2}$$

minimum value of $g(x) = 0 + 2 + 3 = 5$

Now $5 \leq g(x) < \infty$

$$\log_2 5 \leq \log_2 g(x) < \infty$$

$$\Rightarrow a = \log_2 5$$

$$[2^a, 4^a] = [2^{\log_2 5}, 4^{\log_2 5}] = [5, 25]$$

$$\Rightarrow 21 \text{ integers}$$

17. Ans. B

Sol. $\vec{r} \times \vec{a} = \vec{b}$

$$\Rightarrow \vec{a} \times (\vec{r} \times \vec{a}) = \vec{a} \times \vec{b}$$

$$\Rightarrow (\vec{a} \cdot \vec{a})\vec{r} - (\vec{a} \cdot \vec{r})\vec{a} = \vec{a} \times \vec{b}$$

$$\Rightarrow 3\vec{r} - (\vec{a} \cdot \vec{r})\vec{a} = \vec{a} \times \vec{b} \dots (1)$$

Let angle between $\vec{a}$ & $\vec{r}$ is θ

$$|\vec{r} \times \vec{a}| = |\vec{b}|$$

$$|\vec{r}||\vec{a}|\sin \theta = |\vec{b}|$$

$$\sin \theta = \frac{\sqrt{2}}{\sqrt{3}} \Rightarrow \cos \theta = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \vec{a} \cdot \vec{r} = |\vec{a}||\vec{r}|\cos \theta = 1$$

$$\Rightarrow 3\vec{r} = \vec{a} + \vec{a} \times \vec{b}$$

$$\Rightarrow \vec{r} = \frac{1}{3}[\vec{a} + \vec{a} \times \vec{b}]$$

18. Ans. C

Sol. $\lim_{n \rightarrow \infty} \left(\frac{1 + \sin \frac{\pi}{6n}}{1 + \tan \frac{\pi}{7n}} \right)^n$

$$\lim_{n \rightarrow \infty} n \left[\frac{1 + \sin \frac{\pi}{6n}}{1 + \tan \frac{\pi}{7n}} - 1 \right] = \lim_{n \rightarrow \infty} n \left[\frac{\sin \frac{\pi}{6n} - \tan \frac{\pi}{7n}}{1 + \tan \frac{\pi}{7n}} \right]$$

$$= e$$

$$= e$$

$$\lim_{n \rightarrow \infty} \frac{\sin \frac{\pi}{6n} - \tan \frac{\pi}{7n}}{\frac{1}{n}}$$

e

$$\lim_{n \rightarrow \infty} \frac{\cos \frac{\pi}{6n} \left(-\frac{\pi}{6n^2} \right) - \sec^2 \frac{\pi}{7n} \left(-\frac{\pi}{7n^2} \right)}{-\frac{1}{n^2}} = e^{\frac{\pi}{6} - \frac{\pi}{7}} = e^{\frac{\pi}{42}}$$

e

19. Ans. (B)

Use the property :

Focus divides the focal chords into 2 line segments such that their harmonic mean is equal to the semi latus rectum of the parabola.

20. Ans. B

Sol. $np = 8$ & $npq = 4$

$$8q = 4$$

$$q = 1/2 \Rightarrow p = 1/2$$

$$\Rightarrow n = 16$$

$$p(x=r) = {}^{16}C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{16-r} = {}^{16}C_r \left(\frac{1}{2}\right)^{16}$$

$$p(x \leq 2) = p(x=0) + p(x=1) + p(x=2)$$

$$= \left(\frac{1}{2}\right)^{16} [{}^{16}C_0 + {}^{16}C_1 + {}^{16}C_2]$$

$$= \left(\frac{1}{2}\right)^{16} [1 + 16 + 120] = \frac{137}{2^{16}} \Rightarrow k = 137$$

SECTION-II

1. Ans. 28.00

Sol. $\int_0^1 1 \cdot (1-x^4)^7 dx$

$$= \left((1-x^4)^7 x \right)_0^1 - \int_0^1 7(1-x^4)^6 (-4x^3) x dx$$

$$= 0 + 28 \int_0^1 x^4 (1-x^4)^6 dx$$

$$= -28 \int_0^1 (1-x^4)^6 (1-x^4-1) dx$$

$$= -28 \int_0^1 (1-x^4)^7 + 28 \int_0^1 (1-x^4)^6 dx$$

$$\Rightarrow 29 \int_0^1 (1-x^4)^7 = 28 \int_0^1 (1-x^4)^6 dx$$

2. Ans. 0.85 or 0.86

Sol. $y = \frac{\alpha+2}{\alpha-2} \Rightarrow \alpha y - 2y = \alpha + 2$

$$\Rightarrow \alpha(y-1) = 2(y+1) \Rightarrow \alpha = \frac{2(y+1)}{y-1}$$

To find an equation whose roots are

$$\frac{\alpha+2}{\alpha-2} \text{ \& \> } \frac{\beta+2}{\beta-2} \text{ put}$$

$$x = \frac{2(y+1)}{y-1} \Rightarrow x^2 + x + 1 = 0$$

$$\Rightarrow \frac{4(y+1)^2}{(y-1)^2} + \frac{2(y+1)}{y-1} + 1 = 0$$

$$\Rightarrow 4(y^2+1+2y) + 2(y^2-1) + y^2+1-2y = 0$$

$$\Rightarrow 7y^2 + 6y + 3 = 0$$

$$\Rightarrow \left| \frac{\alpha+2}{\alpha-2} + \frac{\beta+2}{\beta-2} \right| = \left| \frac{6}{7} \right| = \frac{6}{7} = 0.85$$

3. Ans. 77.00

Sol. P = both A comes together

Q = both E comes together

$$n(P' \cap Q') = n(P \cup Q)' = n(U) - n(P \cup Q)$$

$$= n(U) - [n(P) + n(Q) - n(P \cap Q)]$$

$$= \frac{9!}{2!2!} - \left[\frac{8!}{2!} + \frac{8!}{2!} - 7! \right]$$

$$= \frac{9!}{2!2!} - [8! - 7!] = \frac{9 \cdot 8 \cdot 7!}{4} - [7 \cdot 7!] = 77.6!$$

4. Ans. 0.82 or 0.83

Sol. $x \tan \frac{y}{x} - y \sec^2 \left(\frac{y}{x} \right) + x \sec^2 \left(\frac{y}{x} \right) \frac{dy}{dx} = 0$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \frac{1}{2} \sin \left(\frac{2y}{x} \right)$$

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$v + x \frac{dv}{dx} = v - \frac{1}{2} \sin 2v$$

$$2 \operatorname{cosec} 2v dv = -\frac{dx}{x}$$

$$\ln(\tan v) = -\ln x + \ln C \Rightarrow x \tan \frac{y}{x} = C$$

$$x = 2, y = \frac{\pi}{4}$$

$$x \tan \frac{y}{x} = C$$

$$2 \tan \frac{\pi}{8} = C$$

$$C = 2(\sqrt{2}-1) = 2(1.414-1) = 0.828 \Rightarrow 0.82 \text{ or } 0.83$$

5. Ans. 0.87 or 0.88

Sol. Let $z = r(\cos \theta + i \sin \theta)$

$$\left| 2z + \frac{1}{z} \right| = 1$$

$$\left| \left(2r + \frac{1}{r} \right) \cos \theta + i \left(2r - \frac{1}{r} \right) \sin \theta \right| = 1$$

$$\left(2r + \frac{1}{r} \right)^2 \cos^2 \theta + \left(2r - \frac{1}{r} \right)^2 \sin^2 \theta = 1$$

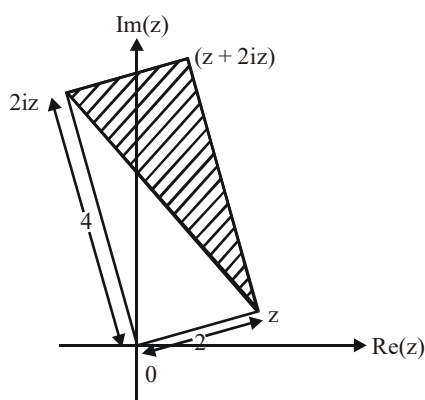
$$\Rightarrow 4r^2 + \frac{1}{r^2} + 3 = 8 \sin^2 \theta$$

$$\text{Now } \frac{4r^2 + \frac{1}{r^2}}{2} \geq 2 \quad (\text{use } AM \geq GM)$$

$$\Rightarrow 4r^2 + \frac{1}{r^2} \geq 4 \Rightarrow 8 \sin^2 \theta \geq 7$$

$$\Rightarrow \sin^2 \theta \geq \frac{7}{8} = 0.87$$

6. Ans. 4.00



obvious from the above figure

$$\text{that area } \frac{1}{2} [2 \times 4] = 4$$

7. Ans. 1.00

Let equation of tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$

$$\text{be } y = mx \pm \sqrt{16m^2 - 9}$$

if it is a tangent of $x^2 + y^2 = 4$

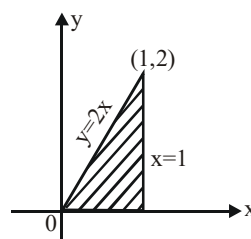
then length of perpendicular from centre = radius

$$\Rightarrow \frac{\sqrt{16m^2 - 9}}{\sqrt{1 + m^2}} = 2$$

$$\Rightarrow m^2 = \frac{13}{12} \Rightarrow [m^2] = 1$$

8. Ans. 1.00

Equation of tangent at origin is $y = 2x$



from figure area bounded is $= \frac{1}{2} \times 1 \times 2 = 1$

9. Ans. 2.00

Area of the parallelogram formed by the lines

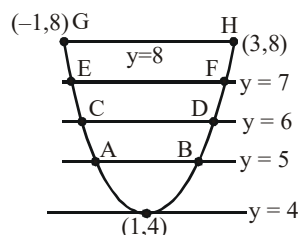
$$y = m_1x + c_1, y = m_1x + c_2$$

$$y = m_2x + d_1, y = m_2x + d_2$$

is given by

$$\left| \frac{(c_1 - c_2)(d_1 - d_2)}{(m_1 - m_2)} \right|$$

10. Ans. 8.00



Rough sketch of the curve $y = x^2 - 2x + 5$ the greatest integer function is discontinuous at $y = 5, 6, 7, 8$ (integral values except $y = 4$)

i.e. the points A, B, C, D, E, F, G, H (in figure)